

Building REST APIs with the LoopBack Framework

LoopBack is an open source Node.js framework, with which you can quickly create REST APIs, and connect devices and browsers to data and services. LoopBack helps to easily create client apps with Android, iOS and AngularJS SDKs, and comes with add-on components for file management, third-party login and OAuth support.



LoopBack is a Node.js framework that is used to design highly-extensible, powerful back-end services for mobile and Web applications. It enables dynamic end-to-end REST APIs with little or zero coding effort through its user-friendly wizard support. It is built on the top of Express.js and adapts most of its functionality like models, routing, middleware, etc, but with minimal coding requirements. It was introduced by StrongLoop and is currently maintained jointly by StrongLoop and IBM. It also comes with good integration support for client-side application development with the available SDKs. This article covers some LoopBack concepts and takes you through the few steps to build a small server application for a weather station.

Installation and building your first application

The current production release is LoopBack v3.0, with long term support (LTS) for v2.x. Install the LoopBack CLI tool, which provides various wizards for generating applications, models, data sources, access control rules, etc, as follows. Choose any LTS versions of Node.js and a compatible npm utility for the current work.

```
npm install loopback-cli -g
```

The IBM API Connect Developer Kit is another solution which uses LoopBack internally. In this article, we'll focus

on the LoopBack CLI tool and v2.x LTS. Let's follow the steps shown below to create a simple app using the application generator.

```
lb app
? What's the name of your application? (osfy-demo)
? Which version of LoopBack would you like to use?
  2.x (long term support)
? What kind of application do you have in mind? api-server
```

Alternatively, you could also choose *hello-world* or *notes* as initial templates. After completing this wizard, all the required modules will be installed in the `node_modules` sub-directory.

Then run the server as follows:

```
node . (or)
npm start
```

...which will prompt the following URLs to the launch home page and a Swagger UI for the REST interface.

```
Web server listening at: http://0.0.0.0:3000
Browse your REST API at: http://0.0.0.0:3000/explorer
```

You may choose any process manager like pm2 for better management of the server.